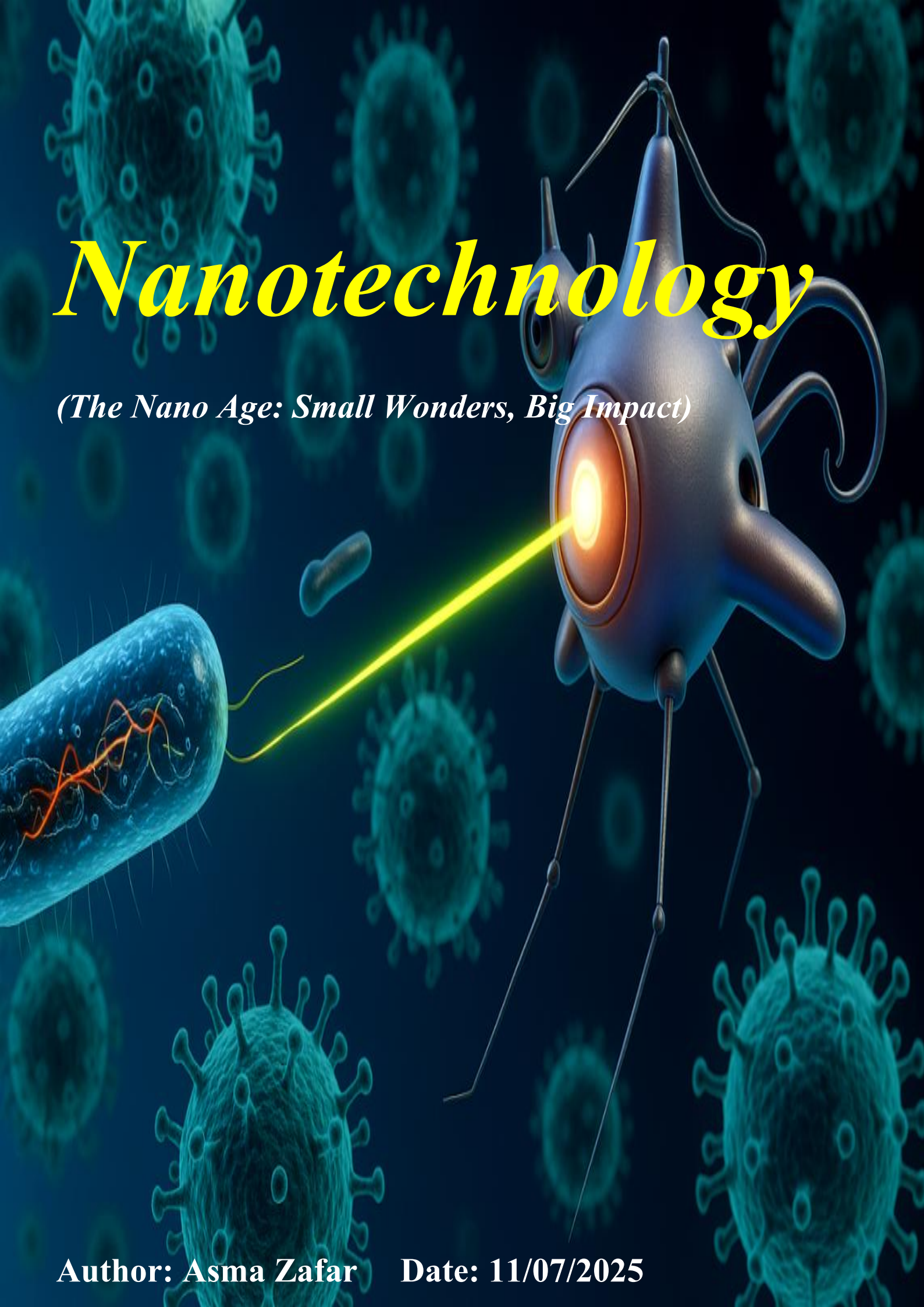


Nanotechnology

A futuristic nanobot with a glowing orange eye and a yellow laser beam, surrounded by various microorganisms like viruses and bacteria.

(The Nano Age: Small Wonders, Big Impact)

The Nano Age: Small Wonders, Big Impact

Definition of Nanotechnology

- ✧ Nanotechnology literally means any technology on a nanoscale that has applications in the real world.
- ✧ Nanotechnology is a group of emerging technologies in which the structure of matter is controlled at the nanometer scale to produce novel materials and devices that have useful and unique properties.
- ✧ Nanotechnology is one of the leading scientific fields today since it combines knowledge from the fields of Physics, Chemistry, Biology, Medicine, Informatics, and Engineering.



Applications of Nanotechnology

Antibacterial Uses

- ✧ For centuries, silver has been used for its ability to destroy bacteria — from ancient Romans treating their water with silver coins to NASA using the metal to purify water aboard the Space Shuttle.
- ✧ Silver (Ag) nanoparticles are embedded in sticking plasters for their ability to inhibit the transmission of viruses.



Cancer Diagnosis and Treatment

Pancreatic cancer has a devastatingly low survival rate (less than 5 percent after 5 years) because it is usually diagnosed at an advanced stage.

Oral Drug Delivery

- ✧ Oral administration of drugs that are currently delivered by injection may be possible in many cases.
- ✧ The drug is encapsulated in a nanoparticle which helps it pass through the stomach to deliver the drug into the bloodstream.
- ✧ Efforts are underway to develop oral administration of several different drugs using a variety of nanoparticles.
- ✧ One company has progressed to the clinical testing stage with a drug for treating systemic fungal diseases.



Biotechnology-Based Medicines

- ✧ Some medicines are made through biotechnological processes, for example those using recombinant DNA (human hepatitis vaccine).
- ✧ Under these processes, the DNA of living creatures (usually bacteria) is altered.
- ✧ Nanotechnology represents a similar approach to the manufacture of pharmaceuticals and other goods.

Nanorobots and Medical Devices

Internal Navigation and Function

- ✧ A navigational network may be installed in the body, which may provide high positional accuracy to all passing nanorobots.
- ✧ This will enable the physician to keep track of the various devices in the body.
- ✧ These nanorobots will be able to distinguish between different cell types by checking their surface antigens.
- ✧ When the task of the nanorobots is accomplished, they can be retrieved by allowing them to exfuse themselves via the usual human excretory channels.

Tumor Treatment and Imaging

- ✧ Mechanical drilling of a small tumor mass by a nanorobot.
- ✧ Improved imaging of the human (or any) body.
- ✧ Nanoprobes (miniature machines) can attach themselves to particles in the body (e.g., antibodies) and emit a magnetic field.
- ✧ Probes that aren't attached to anything don't create a detectable magnetic field.
- ✧ Nano-tracking may be able to detect tumors that are a few cells in size.

Lungs and Dental Applications

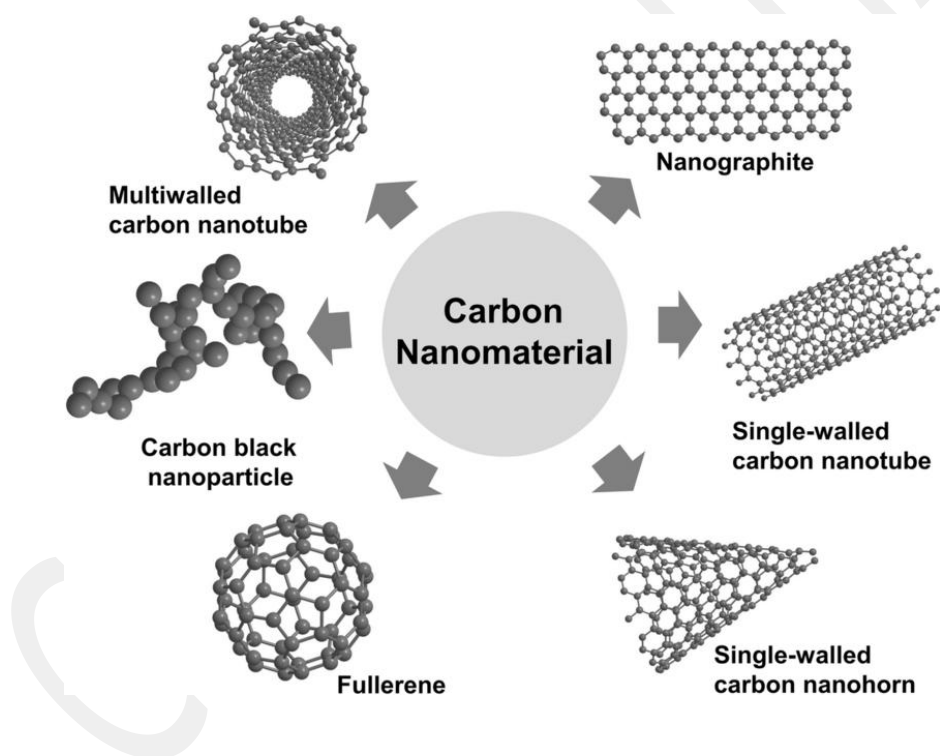
- ✧ A single inhaled nanorobot reaches, deeply inspired into the lungs, enters an alveolar duct, and attaches to the tissue surface.
- ✧ Four remote-controlled nanorobots examine and clean the subocclusal surfaces of a patient's teeth, near the gumline.
- ✧ Medical nanodevices could augment the immune system by finding and disabling unwanted bacteria and viruses.

Carbon-Based Nanomaterials

- ✧ Carbon-based nanomaterials are the second most common nanoscale materials found in consumer products.
- ✧ Properties: lightweight, high strength, electrical and thermal conductivity.

Applications:

- ✧ sports equipment
- ✧ vehicle parts
- ✧ storage of power in batteries
- ✧ moisturizing effectiveness of cosmetics
- ✧ drug delivery



Conclusion & Future Outlook

- ✧ Nanotechnology stands at the forefront of scientific innovation, offering transformative solutions across medicine, materials science, and consumer applications. From silver nanoparticles combating infections to nanorobots

navigating the human body with surgical precision, the field has already begun reshaping how we diagnose, treat, and interact with the world around us. Its interdisciplinary nature—bridging physics, biology, chemistry, and engineering—makes it a powerful tool for solving complex global challenges.

- ✧ Looking ahead, the future of nanotechnology promises even greater breakthroughs. Researchers are developing targeted drug delivery systems, self-healing materials, and next-generation energy storage using advanced nanomaterials like graphene and carbon nanotubes. In healthcare, nanosensors and nanomedicine are expected to enable early disease detection, personalized treatments, and minimally invasive procedures. Meanwhile, industries such as agriculture, electronics, and environmental science are exploring nanotech for precision farming, flexible electronics, and pollution remediation.
- ✧ As we enter deeper into the Nano Age, the convergence of nanotechnology with AI, biotechnology, and quantum computing will likely unlock innovations we can scarcely imagine today. With continued research, ethical oversight, and global collaboration, nanotechnology is poised to become a cornerstone of sustainable progress and human advancement.

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Thank You